

Subject Description Form

Subject Code	AP30016
Subject Title	Materials Science and Analysis
Credit Value	3
Level	3
Pre-requisite/ Co-requisite/ Exclusion	Nil.
Objectives	The aim of this subject is to introduce basic concepts of materials science and experimental techniques for materials analysis.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) categorize materials and describe the differences between materials of different categories; (b) describe the structure of atoms and nature of atomic bondings; (c) describe typical phenomena related to phase change in materials; pursue compositional and structural information of different phases in materials by using equilibrium phase diagrams; (d) describe typical mechanical behaviors in metals and explain the relationships among structure and properties in these materials; (e) apply x-ray diffraction to determine the properties of crystalline materials; and (f) understand the physical principles underlying the magnetic, piezoelectric, and shape memory materials.
Subject Synopsis/ Indicative Syllabus	Materials: classifications of materials. Materials structures: atoms, atomic bonding; crystal structures; unit cells; defects; microstructure; diffusion; phase and phase diagram. Mechanical behaviour: Tensile test; elastic deformation; plastic deformation; dislocation movement; twinning; hardness tests; fracture; impact tests, ductile-brittle transition; creep; fatigue. Techniques for materials characterization: X-ray diffraction; tensile test; thermal analysis of the alloy system;; corrosion of metals; microstructure analysis. Functional Materials and Applications: Electrical properties and applications of superconducting, conducting, semiconducting, dielectric materials; magnetic properties and applications of magnetic materials; applications of ferroelectric, piezoelectric materials and shape memory alloys as smart materials.
Teaching/Learning Methodology	Lecture: Background knowledge behind all experiments will be systematically introduced in lectures. Class work and assignments related to the content of lectures will be used to enhance students learning. Laboratory session: Experiments are essential for students to relate the concepts to practical applications and they are exposed to hand-on experience and proper use of equipment and also analytical skills on interpreting experimental results.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	f
	(1) Continuous assessment	55	√	√	√	√	√	√
	(2) Examination	45	√	√	√	√	√	√
	Total	100						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Assignments will strengthen the students’ basic knowledge and the analytical skills to solve the problems related to this subject. Practical Tests is useful to assess students’ experimental skills and knowledge learned from the lectures and lab works. Examination will review their understanding of the course assess their ability to solve problems. Hence, the proposed assessment methods are necessary to assess the intended learning outcomes.							
	Student Study Effort Expected	Class contact:						
• Lecture				24 h				
• Tutorial				6 h				
• Laboratory				12 h				
Other student study effort:								
• Self-study				78 h				
Total student study effort:				120 h				
Reading List and References	(1) M. F. Ashby, Materials Selection in Mechanical Design, (Pergamon Press, 1992). (2) W.D. Callister, Jr., Materials Science and Engineering: An Introduction (John Wiley and Sons, Inc., 2014)							